ADHAM ELSAYED

AI Engineer | Generative AI, Machine Learning & Computer Vision

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SUMMARY

AI Engineer and final-year Artificial Intelligence Engineering student focused on Generative AI, RAG, and computer vision. Builds Python/FastAPI applications for medical tutoring, evidence-grounded retrieval, and ONNX inference; experienced in model evaluation, REST APIs, SQL, and Power BI.

EDUCATION

Mansoura University **Sep 2023 - Feb 2027 (Expected)**

B.Sc. in Artificial Intelligence Engineering | GPA: 3.13/4.00 | Grade: B+ *Dakahlia, Egypt*

TECHNICAL SKILLS

Generative AI & RAG: LLM APIs, RAG, BGE embeddings, BM25, RRF, ChromaDB, Hugging Face

Model Evaluation & Reliability: Hit@K, Precision@K, MRR, retrieval evaluation, citation validation, evidence gating

Machine Learning & Computer Vision: PyTorch, TensorFlow, YOLO, OpenCV, dlib, face embeddings, ONNX Runtime

Backend & Analytics: Python, FastAPI, REST APIs, SQL, SQLite, Pandas, Power BI, DAX, MLflow

Development & Deployment: Docker, Git, GitHub, Azure AI, Firebase, React, TypeScript, Flutter, C++

PROJECTS

NaFees AI (Anatomy Lab) | GenAI for Medical Education EUI Results

Generative AI, Medical Tutoring, Anatomy Q&A

- Co-designed 3 GenAI tutor workflows for anatomy, medical Q&A, and exam preparation.
- Converted course and professional-exam topics into guided revision and Q&A activities.

MedFlow | Evidence-Grounded Clinical RAG Project Overview

Python, FastAPI, BGE, BM25, RRF, ChromaDB, Groq

- Engineered hybrid retrieval using BGE, BM25, and RRF with 4 evidence-gated response modes.
- Measured 87.5% Hit@4 and 100% citation validity across 7 tests; no fabricated citations.

Smart Basket | Computer Vision & Edge AI GitHub

YOLO, OpenCV, ONNX Runtime, Firebase, Raspberry Pi 5 (planned)

- Trained a 4-class YOLO model; project-reported 96.89% mAP@0.5 and 84.79% mAP@0.5:0.95.
- Implemented ONNX basket tracking with 2-frame confirmation, 3-miss hold, and optional Firebase sync.

Code AI Proctor | AI Exam Review Prototype GitHub

YOLO Classification, FastAPI, PyTorch, SQLAlchemy

- Integrated binary YOLO classification with FastAPI, API-key protection, and instructor-reviewed alerts.
- Connected quiz workflows to PDF-informed learning feedback while preserving human oversight.

Smart Attendance | Face Recognition & Liveness GitHub

React, TypeScript, FastAPI, OpenCV, dlib, SQLite

- Developed exam-scoped face check-in with blink prompts, roster validation, and SQLite storage.
- Verified 24 Python tests and 35/35 browser checks across 7 routes and 5 screen widths.

EXPERIENCE

NaFees AI | GenAI for Education Hackathon **Oct 2026**
AI Engineering Team Member | Medical EdTech *Egypt*

- Scoped the medical-learning prototype with a 5-member AI engineering team.
- Demonstrated tutor workflows and technical feasibility to Startup Track judges.

VOLTIX | Data Analysis Intern **Sep 2026 - Oct 2026**
Data Analytics & Business Intelligence *Egypt*

- Cleaned and validated 1,470 employee records for HR attrition analysis.
- Delivered a Power BI attrition dashboard with DAX across 4 factors: overtime, tenure, pay, satisfaction.

Digitera Program (YIEP 2026) | iCareer **Sep 2026**
Business Track | Case Analysis *Egypt*

- Analyzed business cases and presented recommendations in guided workshops.

Orange Digital Center x CREATIVA Innovation Hubs **Aug 2026**
AI Hackathons Program | Applied AI Prototyping *Egypt*

- Iterated an AI prototype through 40 hackathon hours using mentor feedback.

Digital Egypt Pioneers Initiative (DEPI) **Sep 2025 - Jul 2026**
Microsoft Machine Learning Engineer Track *Remote, Egypt*

- Completed a 10-month Microsoft ML track in NLP, computer vision, evaluation, Azure AI, and MLflow.

ST Smart | Robotics Software Engineering Trainee **Jul 2025 - Sep 2025**
Robotics & Embedded Systems *Mansoura, Egypt*

- Tested 20+ robotics simulations during 50+ hours of C++/Arduino training.

HONORS & AWARDS

2nd Place | Startup Track, GenAI for Education Hackathon 2026 **Oct 2026**
Team NaFees AI (Anatomy Lab) | Egypt University of Informatics

2nd Place | AI Hackathon Mansoura 2026 **Aug 2026**
Team MedFlow | Orange Digital Center Egypt and CREATIVA Innovation Hubs

CERTIFICATIONS

AI & Data Science - Microsoft Machine Learning Engineer | DEPI | Microsoft

Google AI Professional Certificate | Google

Building with the Claude API | Anthropic | Claude Academy

Develop a Generative AI Chat App with Microsoft Foundry | Microsoft

Huawei HCIA-AI Course | Huawei

Technical Enablement Week | GenAI for Education 2026 | Egypt University of Informatics